
COMPUTER CRIME AND PRIVACY

Assignment #2

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PART 1: CYBER CRIME CASE STUDY

SECTION A: REAL-TIME CYBER CRIME CASE

Case Name: MOVEit Transfer Zero-Day Exploitation (May 2023)

In May 2023, a serious security flaw was found in MOVEit Transfer, a file transfer software made by Progress Software. This vulnerability, called CVE-2023-34362, allowed hackers to gain access to computer systems and run commands without proper permission. Within a few days, hackers from different groups started attacking organizations that were using this software, causing one of the biggest data breaches in 2023.

What Happened:

Many organizations around the world were affected by this attack. This included government agencies, hospitals, banks, and large companies. Hackers were able to steal sensitive information from millions of people. The breach showed how quickly a single software problem can cause damage to many organizations.

Laws That Were Used:

In the United States, the Computer Fraud and Abuse Act (CFAA) was the main law used to investigate and prosecute this crime. The CFAA makes it illegal to access computer systems without permission and to cause damage through hacking. People who are convicted under this law can face up to 10 years in prison and pay large fines.

In Europe, the General Data Protection Regulation (GDPR) was applied. This law says that companies must protect personal information and tell people within 72 hours if their data is stolen. Organizations that violated GDPR had to pay fines to European regulators.

In the United Kingdom, the Computer Misuse Act 1990 was used. This law punishes people who access computer systems without permission and cause problems.

SECTION B: SIMILAR CRIME IN PAKISTAN

Case Name: Federal Board of Revenue (FBR) Data Breach (August 2021)

In August 2021, hackers got into the computer systems of the Federal Board of Revenue (FBR), which is Pakistan's tax authority. They were able to steal information about taxpayers, including names, national identity numbers (CNIC), tax IDs, and financial records. The data of over 20 million Pakistanis was exposed in this breach.

Why This Was Serious:

The stolen data could be used by criminals to commit identity theft and fraud. People whose information was leaked could lose money from their bank accounts or have fraudulent loans taken out in their name. This also damaged people's trust in the government because a major government organization failed to protect their personal information.

Laws Used in Pakistan:

The main law used was the Prevention of Electronic Crimes Act (PECA) of 2016. PECA makes it illegal to access computer systems without permission. If someone is caught doing this, they can go to prison for up to 3 years and pay a fine of up to 500,000 rupees. PECA also has a section on cyber-terrorism, which is the most serious crime. This can result in even longer prison sentences and larger fines.

Other laws involved were the Pakistan Telecommunication Act of 1997, which lets the Pakistan Telecommunication Authority (PTA) take action against cyber crimes, and the NADRA Act, which protects information stored in national databases.

What Happened After:

The FBR started investigating the breach internally. The PTA and Federal Investigation Agency (FIA) Cyber Crime Wing also got involved. A case was registered under PECA 2016. Security experts checked the FBR systems to find out what went wrong, and people whose data was stolen were warned to watch their bank accounts and credit.

Comparison Between the Two Cases:

The MOVEit case and the FBR breach were both examples of unauthorized access to data, but they were handled differently. In the international case, the FBI and US government were involved, and the CFAA law was applied. In Pakistan, the case was handled by the FIA under PECA 2016. The international case also involved fines from the European Union because of GDPR violations. The Pakistan case showed that Pakistan's laws for protecting data and punishing cyber criminals are not as strong as those in other countries.

PART 2: PRIVACY INCIDENT CASE STUDY

SECTION A: REAL-TIME PRIVACY INCIDENT

Case Name: Cambridge Analytica Facebook Data Privacy Scandal (March 2018)

In March 2018, it was discovered that Cambridge Analytica, a company that works in politics and elections, had collected personal information about millions of Facebook users without their permission. A programmer named Christopher Wylie revealed that Cambridge Analytica had obtained data on approximately 87 million Facebook users. This information included details about their interests, beliefs, and personal habits. The company used this data to create political ads designed to influence how people voted in elections.

How It Happened:

Cambridge Analytica paid a researcher who created a Facebook quiz app. When people used this app, it collected not just their own information but also data about their friends. Facebook allowed this to happen because their privacy rules at that time were not strict enough. The company also collected data from other sources. This massive amount of personal information was then sold to Cambridge Analytica, which used it for political purposes.

Why This Was a Privacy Problem:

The main issue was that people did not know their data was being collected and used for political campaigning. They thought the information they shared on Facebook was private or only used by Facebook itself. Their personal preferences and beliefs were being studied to influence their political choices. This violates the basic right to privacy and freedom of choice.

Laws Applied Internationally:

In the United States, the Federal Trade Commission (FTC) investigated Facebook and found that the company had broken its own privacy promises. The FTC required Facebook to pay 5 billion dollars in fines, which was one of the largest fines ever given to a technology company. However, there was no single US law like GDPR that covered this situation.

In Europe, GDPR was applied because Cambridge Analytica and Facebook had processed the data of European users without proper permission. GDPR Article 6 requires companies to have a valid reason before collecting personal data. Since Facebook did not have proper consent from users, they violated GDPR. Many European countries fined Facebook and other companies involved.

In the United Kingdom, the Information Commissioner's Office (ICO) found that the companies had broken the Data Protection Act. The ICO fined companies hundreds of thousands of pounds and recommended new laws to protect privacy.

SECTION B: SIMILAR PRIVACY INCIDENT IN PAKISTAN

Case Name: Pakistan Biometric System Data Exposure (2019-2020)

In 2019 and 2020, there were reports that biometric data from Pakistan's national ID system (NADRA) was being sold on the dark web. This included fingerprints, iris scans, and personal information of millions of Pakistani citizens. Unlike the Cambridge Analytica case where data was used for political purposes, this data could be used for identity theft, fraud, and creating fake documents.

How It Happened:

Security researchers discovered that NADRA databases had been accessed by unauthorized people. It was not clear exactly how the breach happened, but it appeared that either the security of the system was weak or someone inside the organization sold the data. The stolen data included biometric information that is very hard to replace or change, unlike passwords.

Laws Applied in Pakistan:

Pakistan did not have a strong privacy law at that time. The main law that could be used was the Prevention of Electronic Crimes Act (PECA) 2016, which makes it illegal to access information systems without permission. However, PECA focuses on punishing the hacker, not on protecting privacy rights. The NADRA Act requires security measures for personal data, but it does not have strict penalties for violations.

There were no specific laws about biometric data protection or laws requiring companies to notify people when their data is stolen. The FIA investigated the case under PECA 2016, but because of weak privacy laws in Pakistan, the government could not do much to protect the affected people or force NADRA to improve security.

People whose biometric data was stolen had no real way to get compensation or change their fingerprints and iris scans. The incident highlighted the need for stronger privacy protections in Pakistan.

Comparison Between the Two Cases:

The Facebook Cambridge Analytica case and the Pakistan biometric data case both involved misuse of personal information, but in different ways. In the Facebook case, data was collected without proper consent and used for political purposes. In the Pakistan case, sensitive biometric data was stolen, likely due to poor security. The Facebook case resulted in large fines and new regulations in Europe and stricter privacy rules by Facebook. The Pakistan case showed that the country did not have adequate laws to protect privacy or punish companies that fail to secure personal data.

SECTION C: SUGGESTIONS FOR NEW LAWS IN PAKISTAN

Current Situation:

Pakistan does not have a comprehensive privacy law like GDPR in Europe or strong data protection regulations like many developed countries. The laws that exist, such as PECA 2016, focus more on punishing hackers than on protecting people's privacy rights. This has led to several problems including data breaches and misuse of personal information.

Recommended Changes to Protect Privacy:

1. Create a Strong Data Protection Law. Pakistan should pass a comprehensive Data Protection Act that clearly defines how companies can collect, store, and use personal information. This law should require companies to get explicit consent from people before collecting their data. It should also set clear rules about how long data can be kept and when it must be deleted.
2. Establish a Data Protection Authority. Pakistan should create an independent authority similar to the European Data Protection Board to monitor companies and enforce privacy laws. This authority should have the power to investigate complaints, conduct audits, and fine companies that violate privacy rights.

3. **Require Data Breach Notification.** Companies should be legally required to inform people within a specific time period if their personal data is stolen or exposed. Currently, there is no such requirement in Pakistan, so many people never know their data has been compromised.
4. **Strengthen Penalties for Privacy Violations.** Current laws have relatively small fines and short prison sentences for privacy violations. Pakistan should increase these penalties to make companies take privacy seriously. Fines should be large enough to significantly impact the company's finances.
5. **Protect Specific Types of Data.** Laws should provide extra protection for sensitive data like biometric information, financial records, and health information. These types of data need stronger security requirements than regular personal information.
6. **Give People Rights Over Their Data.** People should have the right to know what data is being collected about them, the right to correct wrong information, and the right to have their data deleted when they no longer want the company to have it.
7. **Regular Security Audits.** Government should require regular security checks of organizations that handle personal data. This would help find problems before data is stolen.
8. **International Cooperation.** Pakistan should work with other countries and international organizations to set standards for privacy protection. This would help Pakistani companies understand global expectations and make it easier to do business internationally.

Why These Changes Are Important: With more people using the internet and more companies collecting personal information, privacy protection has become very important. Pakistan has experienced several major data breaches, and without strong privacy laws, people have no protection. These new laws would give people control over their personal information and encourage companies to invest in better security.

CONCLUSION

This report looked at both cyber crimes and privacy violations through real cases. The MOVEit case showed how a technical problem can lead to a major breach, while the FBR case showed that Pakistan has laws to punish cyber criminals but needs to apply them more effectively. The Cambridge Analytica case showed how privacy can be violated even when no illegal hacking occurs, while the Pakistan biometric case showed that Pakistan lacks adequate privacy laws.

The main difference between cyber crime and privacy violation is that cyber crime usually involves illegal access to systems, while privacy violations involve misuse of data that may or may not have been legally obtained. Pakistan has made progress with PECA 2016, but needs to develop a comprehensive privacy law similar to GDPR.

Going forward, Pakistan should focus on creating stronger privacy laws, improving data security in government systems, and making sure that companies take privacy seriously. This would help protect the personal information of millions of Pakistanis and encourage innovation in technology while keeping people safe.

Screenshots of the prompts used for research :

What is cyber crime? Give me some examples of major cyber crimes that happened in the last few years.

Tell me about a major data breach or cyber attack that happened in 2023. What was affected and how many people were impacted?

What are the main laws in the US that are used to prosecute cyber criminals?
What is the Computer Fraud and Abuse Act?

What is GDPR and how does it apply to data breaches in Europe?

What laws does Pakistan have to deal with cyber crimes? Is there anything like PECA or other cyber crime laws?

Have there been any major data breaches of Pakistani government databases? What happened and which agencies were involved?

Does Pakistan have strong privacy laws? What happens if a company in Pakistan leaks people's personal information?

Compare cyber crime and privacy violations. What is the difference between them